

# ENVIRONMENT AND FEEDING RELATIONSHIPS

Earlier you have studied that there are three types of environment i.e. land, water and air. Grasslands, wetlands and forests are an example of terrestrial environments, whereas lakes, rivers and oceans belong to the aquatic environment. These environments have different characteristics and features by which they are recognized as well. They are the habitats of many living organisms. Have you ever thought, why all plants and animals are important to

## In this Chapter you will learn about:

- Ecosystem
- Habitat
- Kinds of Habitat
- Biotic components and their relationship with food chains and food web

### All the students will be able to:

- ✓ Explain the ecosystem
- ✓ Define the term habitat
- ✓ Compare the different kinds of habitats
- ✓ Investigate the various features that allow animals and plants to live in a particular habitat
- ✓ Identify the factors that cause daily and yearly changes in a habitat
- ✓ Explain how living things adapt to daily and yearly changes in their habitat
- ✓ Explain the ways in which living things respond to changes in daily environmental conditions such as light intensity, temperature and rainfall
- ✓ Explain why food chains always begin with a producer
- ✓ Illustrate the relationship between producers and consumers
- ✓ Describe two food chains in the environment around them
- ✓ Explain a food web

the environment? Why some organisms are abundant in a particular environment? How do organisms find the best place for them to live? Why survival of species at some places is easier than other places?

You have studied that the community of living organisms (biotic component) interacts with non-living environment for food, shelter and protection. The interaction of community with non-living environment (abiotic factor) in a particular area for survival is called **ecosystem**. Let us explore different types of ecosystem and interaction present in them.

## ECOSYSTEM

✓ Explain the Ecosystem.

Ecosystem is a place in water or at land where biotic component i.e. community interact with non-living environment. The community of biotic components such as plants, animals, and microorganisms and abiotic components (water, air, nutrients and solar energy) when interacts for survival in the environment make up an **ecosystem**. These fresh water, terrestrial or ocean **ecosystems** could be as large as the desert, forest and it can be as small as a pond or school garden. All those components that interact in an ecosystem are called factors. As we have studied earlier, there are of two types; Biotic and Abiotic factors.

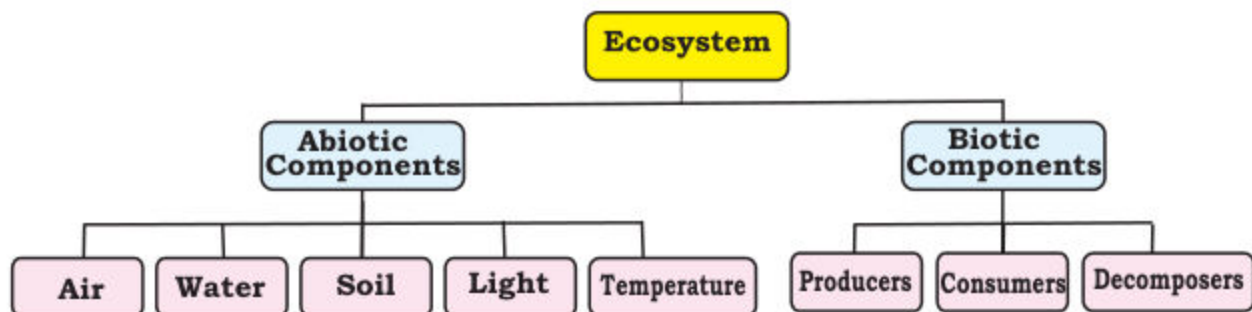


Figure: 4.1 Components of an ecosystem

When different organisms (biotic factors) interact with members of their own species as well as with the members of other species and with their physical environment (abiotic factor), they form an ecological system or ecosystem. This interaction may be for the sake of food, shelter, protection and reproduction. The biotic and abiotic components are very important for sustaining an environment.

**Activity 4.1:** Observe the ecosystem and make a list of biotic and abiotic components.



Pond ecosystem



Grassland ecosystem.



## Pond Ecosystem

## Grassland Ecosystem

The interactions between biotic and abiotic components in ecosystems help in maintaining the balance between these components and the environment. These interactions are responsible for obtaining the stability in the environment.

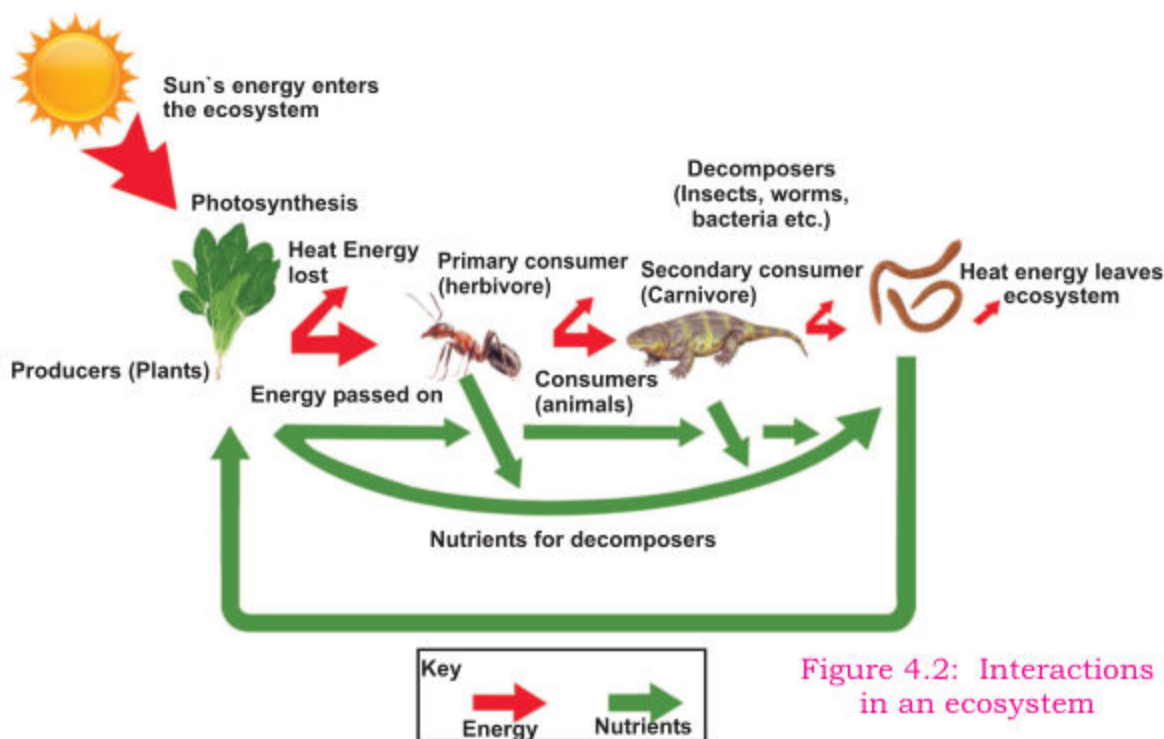


Figure 4.2: Interactions in an ecosystem

The deserts, seashore, mountains, rivers, oceans, grass lands and rainforests are some of the world's ecosystems. Earth itself is a huge ecosystem. Ecosystem can be defined in summarized form as:

**Ecosystem = Habitat + Community of living things**

The Earth has many different ecosystems, which vary in life forms, temperature, moisture, light, and other factors. Each of these ecosystems has distinct life forms living in it, which form complex communities of interdependent organisms.

## HABITAT

✓ Define the term habitat.

A habitat is an ecological area or environment in which species of animals, plants and all types of organisms live. The term typically refers to the space in which the organism lives and where it can find food, shelter, protection and mates for reproduction. The habitat is a place of living which provides the organisms all requirements to survive. Habitat is not necessarily a geographic area; for example, for a parasite, the habitat is the body of its host where from it gets food and find suitable environment to live in. Habitat is made up of physical factors (abiotic) such as kind of soil, moisture in air, range of temperature, availability of light throughout the year and some biological (biotic) factors like availability of food.



Figure 4.3: Different habitats

## KINDS OF HABITATS

✓ Compare the different kinds of habitats

According to its physical characteristics, there are two types of habitats:

(1) Aquatic and

(2) Terrestrial Habitat



## Aquatic Habitat:

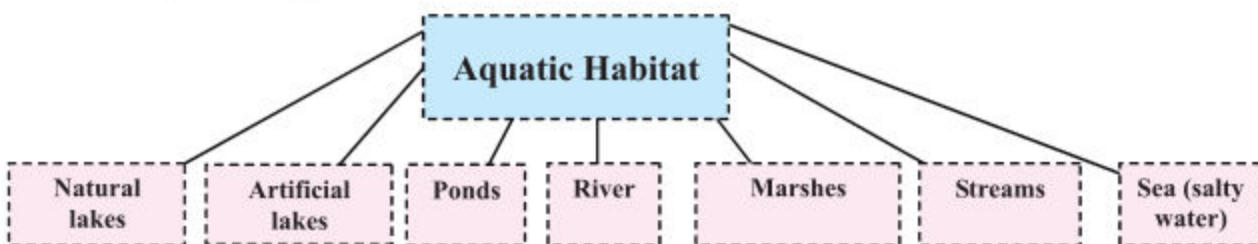
The living organisms which live in aquatic environment surrounded by water, they get all the resources from the water in which they live. They develop their body organs to survive in this environment. The aquatic habitat also varies with the nature and physical state of water e.g:

Freshwater Habitat  $\Rightarrow$  Water Contain very low amount of salt

Marine water Habitat  $\Rightarrow$  Water Contain high amount of salt

Estuarine water Habitat  $\Rightarrow$  Area where river meet with sea.

The examples of fresh water and flowing water habitat are: rivers, streams, lakes, ponds and pools.



## Rivers and Streams:

Rivers and streams refer to systems with rapid flowing waters that move in a unidirectional way. For several species rivers and streams, are safe places to live. Crustaceans like crayfish and crabs; and mollusks such as clams and limpets are commonly found in streams and rivers. Various mammals such as beavers, otters and river dolphins also inhabit rivers and streams ecosystems.



Figure 4.4 River Indus

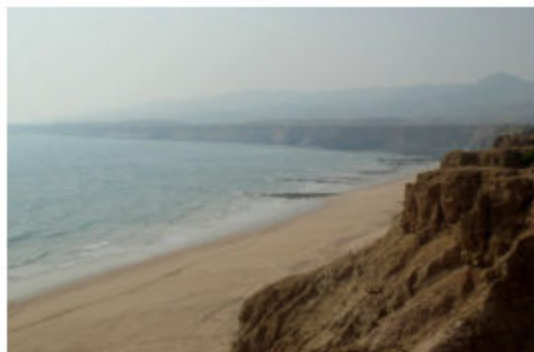


Figure 4.5 Clifton Beach

## Lakes, ponds and pools:

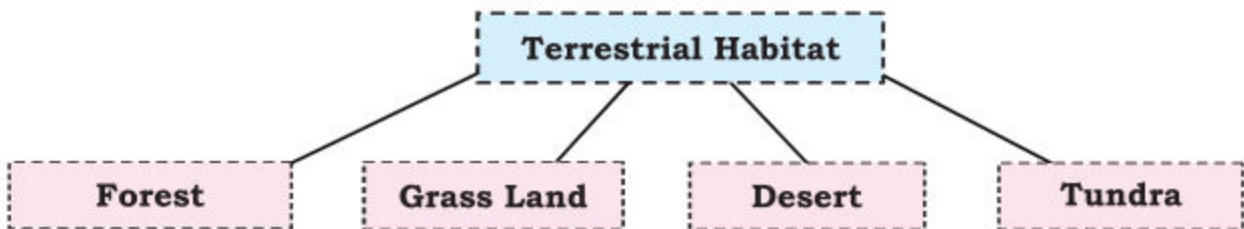
Ecosystems of still waters such as lakes and ponds that have a community of biotic (living organisms) and abiotic (physical objects) interactions. Ponds and lakes have a diverse variety of organisms including algae, rooted and floating-leaved plants, invertebrates such as crabs, shrimps, crayfish, clams etc, amphibians such as frogs and salamanders; and reptiles like alligators and turtle.

## Terrestrial Habitat:

Organisms living on land obtain their resources from soil, land and air. Factors which influence life on land are light, temperature, rain, availability of water etc. Following are the main types of terrestrial habitat.



Figure 4.6 Photographs of Terrestrial Habitat



### Forest:

Forest is land where rainfall is high i.e 250 to 450 mm annually. It has many trees and combination of herbs, shrubs, plants, seedlings and several varieties of birds, mammals, insects, reptiles, amphibians and small creatures.

### Grassland:

Habitat where rainfall is lower than forest so it does not support growth of trees. The condition in grassland grows i.e wheat, maize, long grasses which are the food source for grazing animals. The grassland further divided into two main types:



**i) Temperate Grassland:** Temperate grass and has a mild climate and four seasons. Soil of these grasslands is rich in humus.

**ii) Tropical Grassland:** Tropical grasslands are near the equator and the environment remains warm all around the year. Rainy and dry seasons are found here. Trees are abundant and soil is poor.

### **Desert:**

Area where rain fall is very low resulting barren area where living condition is hostile for plants and animals. Largest dessert in Pakistan is Thar.



Figure 4.7 Desert

### **Tundra:**

Area below ice caps which are very cold regions, it is difficult to live in these areas, only migratory animals can survive under favorable conditions i.e Antarctica. The organism like Polar Bear, Arctic Foxes, Snowy owl are the inhabitant of this biome.



Figure 4.8 Tundra

**Activity: 4.2:** Sorting Animals and Plants according to their Habitats:

### **What do I need:**

List of animals and plants (given below in the box).

### **What to do:**

Look at the habitats in the table below. Names of different plants and animals are given in the box below the table. Copy the table in your exercise book and write the name of each plant and animal in the correct column.

#### **INVESTIGATE**

- Which place is most suitable for growing coconut trees?
- Why pine trees grow in hilly areas?
- Why Mango trees cannot grow in Gilgit?
- Why polar bears are not found in Thar Desert?

| Sea/Deep water | Forest/wood | Garden | Urban Town | Desert |
|----------------|-------------|--------|------------|--------|
|                |             |        |            |        |
|                |             |        |            |        |
|                |             |        |            |        |

Snail, flower, seaweed, camel, ladybird, squirrel, owl, deer, pegeon, rat, oak tree, shark, earth worm, cactus.

**ACTIVITY 4.3:** Comparing the different kinds of Habitat: Group Activity (Poster Presentation)

### What I need:

Textbook, Internet/ old magazines and books

### What to do:

- Read the text given in your textbook about different types of habitat.
- Search the internet /read books from the library/ magazines/old books to find out the qualities of each habitat.
- Visit different habitats (if possible).
- Based on readings and observations compare different kinds of habitat.
- Discuss with your teacher, classmates, senior classes and parents.
- If possible interview with an environmentalist.
- Present your findings to the whole class in the form of poster presentation.

## DIFFERENT FEATURES THAT ALLOW ANIMALS AND PLANTS TO LIVE IN A PARTICULAR ENVIRONMENT

- ✓ Investigate the various features that allow animals and plants to live in a particular habitat.

Each kind of environment has different kinds of animals and plants habitat. Animals and plants develop many ways to survive in a particular environment. Palm trees grow near coastal areas and pines at high altitude.



Polar bear lives in cold area and lion in grassland because their requirements of life are present in these areas. If a particular type of need do not meet by a habitat, the living organisms will either migrate to better place or adopt themselves according to the habitat. For example human beings living in warm area produce black pigments in their skin to protect them from sunlight but the people living at high altitude do not produce black pigment in very high quantity so their complexion remains fair.

### **How organisms adapt to live in a particular habitat?**

Group of organisms of a particular type live in a specific habitat called population. They have special features to survive in that habitat. Plants require sunlight, water and minerals from soil to prepare their food by the process of photosynthesis. In order to receive larger amount of light most of the plants adopted to produce broader leaves. If due to some geographical changes there is water scarcity the plant become unable to survive in this condition, therefore they develop some characteristics by which they reduce water loss for example: cactus plant adopted spines on its stem. Some animals like earth worm prefer to stay in the dark hence they burrow under the soil. This adjustment of living organisms in a particular environment for the usage of available resources is called adaptation.

Different factors, such as duration of light, temperature, availability of water, speed of wind, humidity in air, snow, salty water, minerals in soil, availability and nature of food, and protection from predators influence living organisms to modify their characteristics.

**Activity 4.4:** Name one organism in which you can find adaptation for each of the above mentioned factors.

Climate change is a change in average weather conditions. Climate change is caused by factors such as biotic processes, variations in solar radiation received by Earth, tectonic plates and volcanic eruptions. Certain human activities have also been identified as significant causes of recent climate change, often referred to as "global warming".

Heat received from the sun greatly influences the temperature of a place. Most of the living organisms are active at temperature in between  $0^{\circ}\text{C}$  to  $45^{\circ}\text{C}$ . Organisms living at below  $0^{\circ}\text{C}$  temperature have adopted the fur on their skins. Shedding leaves and hibernation (winter sleep under the mud) are two adaptive processes which have enabled plants and animals to survive under minimum temperature.

In some habitats due to scarcity of water animals and plants adapted to store water and reduce water loss. For example the cactus plant has spines instead of leaves to overcome the water loss. In animals, camels drink lot of water at one time to store it in their humps.

### **ENVIRONMENTAL FACTORS THAT CAUSE DAILY AND YEARLY CHANGES:**

- ✓ Identify the factors that cause daily and yearly changes in a habitat.

As you know that environmental factors are not constant and keep on changing in different parts of a habitat, at different times of a day or at different times of the year. If you note the temperature of a place at different times of a day, you will find it different. Similarly, sun shines during the day so days are warm and bright.

#### **DO YOU KNOW?**

Habitat is not necessarily a geographic area e.g. for a parasite habitat is the body of its host where from it get food and found suitable environment to live

However at night when the sun does not shine on our part of the Earth it is dark and colder. Some animals are most active during the daytime. They are diurnal and some are most active during the night, they are nocturnal. Can you give some examples of diurnal and nocturnal animals found at the place where you live?

### **Seasonal or yearly changes in Environment living things adapt to daily and yearly changes:**

There are four seasons in a year. In every season environmental factors change a lot. In summer, nights are short and days are long.



It is quite warm during the day but even after sun set it does not get very cold. In winter, days are short and nights are long. In Sindh it does not get very cold in the day and at night. However in Punjab, Khaybar Pakhtoon khwa and Balochistan it is comparatively cold. There may be frost and snow.

Natural disasters such as drought, floods, earthquakes also bring changes in environment. If there is no rain for a long time then this period is called drought. During drought most plants and animals found on land in lakes, ponds and streams will die. Some animals move to other places.

Flood is another disaster caused by heavy rains for longer time. During a flood many plants, animals and human beings die or move to another safer places. Lightning also cause fire in forest, which burn different animals and plants. It takes many years for a forest to grow back.

Earthquakes also bring changes in the environment. On October 08, 2005 and recently on October 26, 2015 a large spread area of Pakistan was damaged due to massive earthquakes.

#### **Activity 4.5:** Investigating a local ecosystem

##### **What I need:**

- Choose a local ecosystem to study.
- Pen/Pencil
- Exercise book

##### **What to do:**

Select an ecosystem near your school.

- Visit the selected ecosystem, observe and record observations in exercise book.
- Present observation and conclusions to the whole class.

##### **What I observed:**

| Name of producer | Name of primary consumer | Name of secondary consumer | Name of tertiary consumer | Name of non living or abiotic factors |
|------------------|--------------------------|----------------------------|---------------------------|---------------------------------------|
|                  |                          |                            |                           |                                       |
|                  |                          |                            |                           |                                       |
|                  |                          |                            |                           |                                       |

##### **INVESTIGATE**

Investigate a few organisms that live in a

1. Brightly lit environment
2. Shady environment
3. Dark environment

Write your observations in your exercise book. Consider the following questions for investigation.

Questions for discussion:

- What do these animals and plants eat?
- Does this place have more producers than the consumers?
- How many food chains are present in this ecosystem?
- Did you observe any decomposer?
- Draw a web of ecosystem make sure to include the sun.

**What I Conclude:**

**Activity 4.6:** Comparing environmental factors in day time and at night time.

**What I need:**

- Thermometer for measuring temperature.
- Anemometer for measuring the speed of wind.

**What to do:**

Place of observation:- \_\_\_\_\_ Date: \_\_\_\_\_

| Environmental factor              | Day time | Night time |
|-----------------------------------|----------|------------|
| Temperature in $^{\circ}\text{C}$ |          |            |
| Light Intensity                   |          |            |
| Wind Speed                        |          |            |

Which animals are found in day time and which are found at night?



## What I Conclude:

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In the above examples of ecosystem you can observe a particular environment in which particular types of living organisms live. For example you cannot find tall wooden trees in grasslands or in a pond you cannot find large whales or sharks you will find only small fishes and frogs.

## How living things adapt to daily and yearly changes in their habitat?

Each kind (species) of living organisms is able to live in a particular habitat. Each has special features which help it to survive in that habitat. These special features are called adaptations. The adaptations help the organism to cope with the environmental factors in their habitat. We say that each specie is adapted to its habitat. Organisms that are not well adapted to their habitat may not survive. Examples of these adaptations are camouflage, migration, hibernation, estivation, body covering etc.

### Camouflage:

Some animals have the property to change their colour according to their surrounding in order to hide from enemies for their survival.

### Migration:

It is the relatively long distance movement of animals from one place to another, usually on seasonal basis. Usually birds, fish, reptiles etc migrate to protect themselves from extreme hot and extreme cold temperatures.

### Hibernation:

It is a winter sleep that helps animals to save energy and survive in the winter without eating much.

## **Estivation:**

When animals slow their activity in hot and dry summer months. During a period of Estivation, many reptiles go underground where it's cooler. Estivation is very similar to hibernation, when some mammals spend the winter moving very little and sleeping a lot, in order to save energy.

## **Body coverings:**

It help animals to survive with environmental changes. For example scales are a type of body covering that fish and reptiles have. Fish are animals that live in water. Reptiles are cold-blooded animals. This means that they can not maintain their body temperature by themselves but we can maintain our temperature in variable temperature.

## **THE PHYSICAL ENVIRONMENT**

- ✓ Explain the ways in which living things respond to changes in daily environmental conditions such as light intensity, temperature and rainfall.

The climate of a place is very important in the physical environment of a place. It determines the type of organisms living there. As the environment can never be best suited for every organism living in it, some organisms have adapted to their environment by developing certain features which enable them to live successfully in their environment. Following are the physical factors which make up the physical environment.

### **Light:**

Plants use light energy from the sun to make their own food. Light intensity and quality affect photosynthesis. As other forms of energy depends directly or indirectly on green plants for food, they are also dependent on light for survival. Light also enables many living organisms to see, so that they can move about, find food and detect danger.



Some organisms such as earthworm prefer to stay in the dark. They have special features to live in a dark environment. Insects such as fireflies and deep sea fish produce their own light to illuminate their prey, confuse their predators and attract their mates. Visibility affect detection of prey.

### **Temperature:**

The heat received from the Sun greatly influences the temperature of a place. Most organisms are active at temperatures between  $0^{\circ}\text{C}$  and  $45^{\circ}\text{C}$ . However, aquatic animals and marine plants can remain active at freezing temperature of around  $-1^{\circ}\text{C}$ . On the other hand certain invertebrates live in hot springs which have temperatures about  $59^{\circ}\text{C}$  while some algae can grow in water having temperature of up to  $80^{\circ}\text{C}$ .

A particular specie has a level of tolerance range within which it functions best. This is its optimal range for that factor within which it functions best, it may be broad or narrow depending on the species e.g. experiencing prolonged period of extremes of temperature that can result in heat stroke or hyperthermia. Some plants are shade loving while others require a high light intensity if they are to thrive.

### **Humidity:**

Humidity affects water loss from plants and evaporation from animals and human beings.

### **Wind speed:**

Wind speed can cause physical damage to plants and affects rate of transpiration; can contribute to dehydration in animals and can affects behavior. Stillness of air can be significant.

### **Soil or water nutrients, salinity:**

Terrestrial plants have particular nutrient requirements.

### **Rainfall:**

Rainfall contribute to water availability.

## FOOD CHAIN

- ✓ Explain why food chains always begin with a producer.
- ✓ Describe two food chains in the environment around them

Organisms which live in a particular habitat depend on each other for their food to get their energy requirements. This transfer of energy by eating is called food chain. Food chain has biotic factors like producer, consumer, and decomposer. It always starts from producers which are sun and plants. Sun is the main component of this process because without the Sun, plants (producers) cannot make food (photosynthesis) and transfer of energy by eating will become impossible.

Thus Plants make their own food by using energy from the sun, therefore, plants are producers. Animals cannot make their own food therefore, they get energy by eating plants and animals, they are called consumers. Cows, sheep, goats, deer etc. are grazing animals; they obtain their energy by eating plants or producers so they are called primary consumers. Most humans and some animals get their energy by eating meat of these grazing animals (primary consumers) therefore they are secondary consumers. Similarly secondary consumers when eaten up by other animals then they become tertiary consumers. Food chain always starts with Sun and producers (green plants) because they prepare and provide food to herbivores. Herbivorous which feed on plants are called primary consumers and carnivorous which eat primary consumers are secondary consumers, when secondary consumers are eaten up by other animals they become tertiary consumer in a food chain.

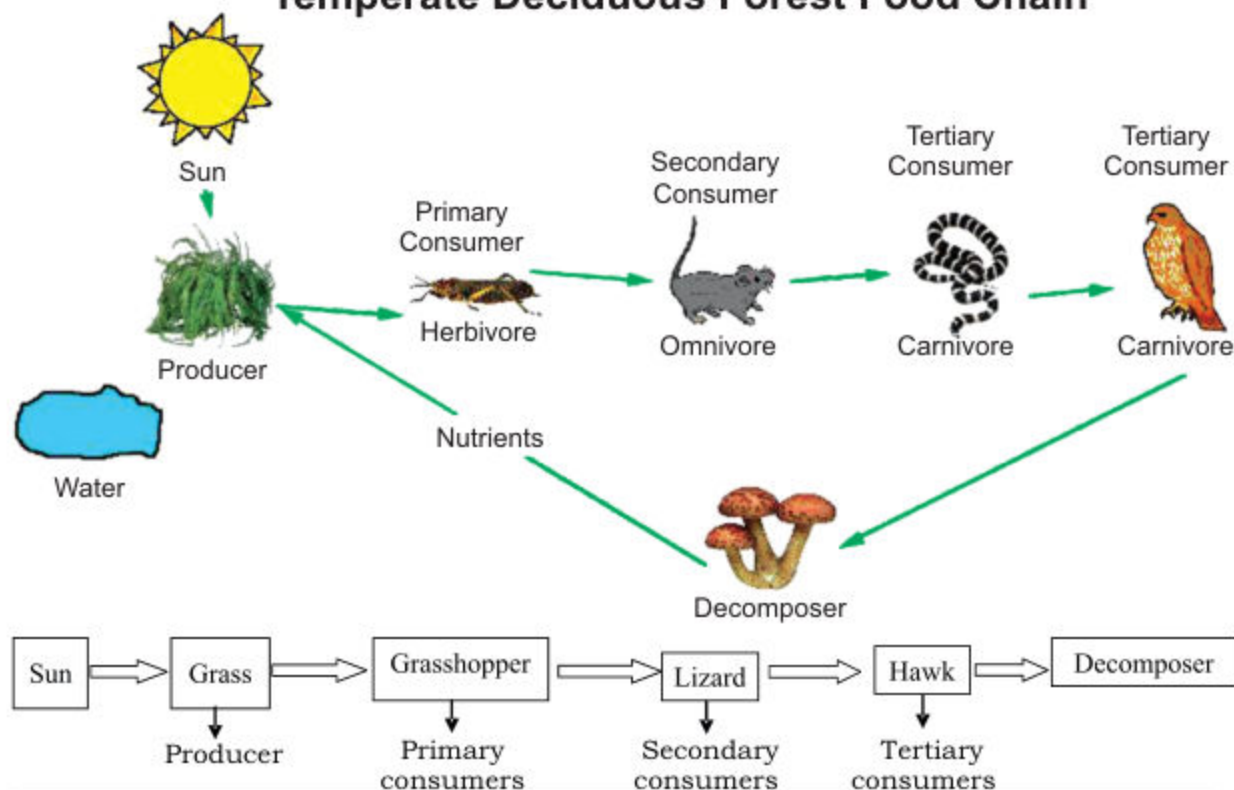
The bodies of dead animals and plants start rotting. This process of rotting is called decomposition. Many fungi and bacteria help in the process of decomposition and are called decomposers. This process of transfer of energy from producer to consumers and then to decomposers is known as a food chain.



## FOREST FOOD CHAIN

✓ Illustrate the relationship between producers and consumers.

### Temperate Deciduous Forest Food Chain



**Activity 4.7:** Identifying abiotic and biotic factors in a pond food chain by observing given picture.

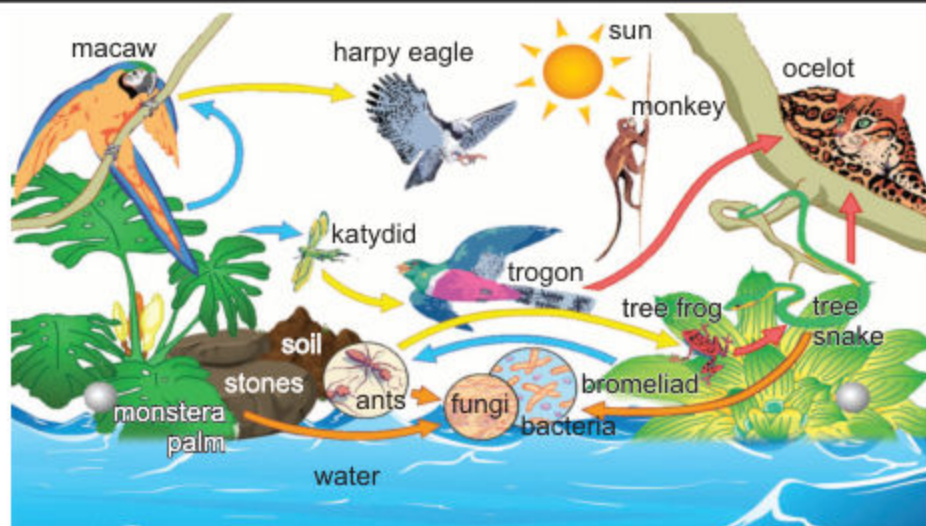


Figure 4.9 Pond food chain

### What to do :

- Observe the given drawing of an ecosystem of pond and fill the table given below.
- If possible, teacher can also arrange a visit to observe river/sea and facilitate students to fill the similar table.
- Students can also visit with their parents and share their findings with the whole class.

| Biotic components | Abiotic components |
|-------------------|--------------------|
|                   |                    |

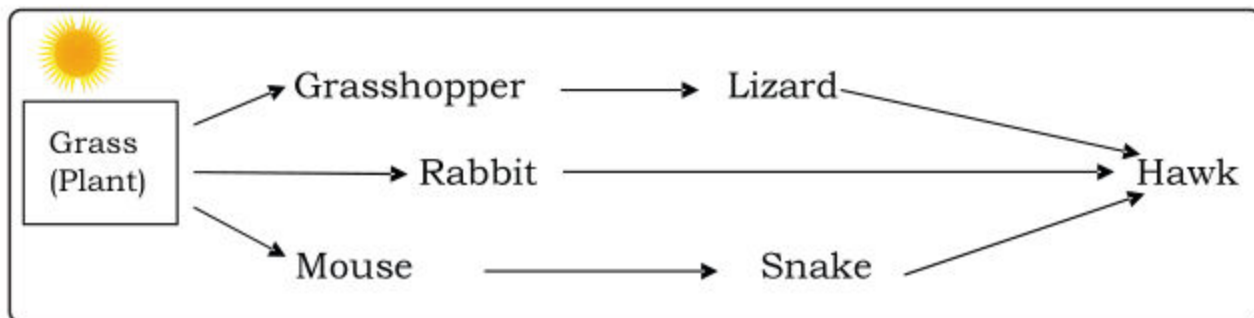
#### Activity 4.8: Working in a group of 3-5.

Draw 2 food chains found around you. Compare your food chain with the other groups and discuss the reasons of difference in each food chain.

### FOOD WEB

✓ Explain a food web.

In nature simple food chains occurs rarely. The same organisms may operate in the ecosystem at more than one trophic level, i.e it may derive its food from more than one source. Even the same organisms may be eaten by several organisms of a high trophic level or organisms may feed upon several different kinds of organisms of a lower trophic level. Thus in an ecosystem various food chains are linked together and intersect with each other to form a complex network called food web.



Food web



Above diagram indicates that an organism does not depend solely on another. The resources of habitat are shared specially in the beginning of food chain. The grass is eaten by a variety of animals such as grasshopper, rabbit and mouse and some of the animals are consumed by several predators. The rabbit in turn directly eaten by hawk or other predator like leopards etc. and their dead bodies are decomposed by decomposers.

In nature many alternatives are found which all together constitute some sort of inter linking pattern of the food web.

## Summary

- When organisms (biotic factor) interact with non living things (abiotic factor) in an environment they form ecological system or ecosystem.
- Natural home of an organism where it lives and reproduces is its habitat.
- Plants and animals adapt to live in a particular place.
- Light intensity, temperature, water, droughts, floods, earthquakes, etc. are the factors that can bring changes in a habitat.
- Human beings are also bringing changes by polluting the environment.
- Green plants produce food through photosynthesis, therefore they are called producers. Animals eat plants so they are consumers.
- The feeding relationship among organisms is called a food chain.
- Food chain always begin with a producer, which produce food in the presence of sunlight.
- In an ecosystem several food chains overlap to form a network called food web.

## Review Exercises

### 1. Answer the following questions:

- i) What is habitat? Name different types of habitat and write their distinct features.
- ii) How living organisms adapt in changing environment?
- iii) Define following terms:
  - a) An ecosystem      b) Population      c) Community
- iv) Describe factors that can bring daily and yearly changes in the habitat.
- v) Describe adaptations of some aquatic animals to live in their habitat.
  - a) List the various features that allow animals and plants to live in a particular habitat
  - b) Explain a food chain and a food web and give at least two examples of each.
  - c) How people change habitat? What can be done to protect habitat?

### 2. Encircle the correct answer in the following :

- i) The interaction of living organisms with each other and with their environment is:
  - (a) Ecology    (b) Ecosystem    (c) Habitat    (d) Food Chain
- ii) A place of living according to particular environment of living thing is:
  - (a) Ecosystem                      (b) Habitat
  - (c) Temperature                  (d) Ecosphere
- iii) Area where river meet with sea is:
  - (a) Freshwater                      (b) Marine water
  - (c) Estuarine water                  (d) Stationary water
- iv) Area below ice cap have very cold climate:
  - (a) Forest                              (b) Desert
  - (c) Grass land                          (d) Tundra
- v) The food chain always starts from:
  - (a) Primary consumer              (b) Secondary consumer
  - (c) Producer                          (d) Decomposer